

Johansen Cointegration Test

Date: 07/09/26 Time: 11:57
Sample: 1/04/2010 12/30/2025
Included observations: 4159
Lags interval (in first differences): 1 to 4
Endogenous variables: EURUSD CADUSD
Deterministic assumptions: Case 3 (Johansen-Hendry-Juselius): Cointegrating relationship includes a constant. Short-run dynamics include a constant.

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.** Critical Value
None *	0.003255	15.88250	15.49471	0.0437
At most 1	0.000562	2.337245	3.841465	0.1263

Trace test indicates 1 cointegrating equation(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Max-eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.003255	13.54526	14.26460	0.0647
At most 1	0.000562	2.337245	3.841465	0.1263

Max-eigenvalue test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegrating Coefficients (norma

EURUSD	CADUSD
19.65933	-17.05358
-3.517003	12.74703

Unrestricted Adjustment Coefficients (alpha):

D(EURUSD)	-0.000291	-9.33E-05
D(CADUSD)	3.49E-05	-9.32E-05

1 Cointegrating Equation
Log-Likelihood: 32679.10

Normalized cointegrating coeffici

EURUSD	CADUSD
1.000000	-0.867455
	(0.13197)

Adjustment coefficients (standard

D(EURUSD)	-0.005730	(0.00197)
D(CADUSD)	0.000687	(0.00121)